GH raisoni INTERNATIONAL SKILLTECH UNIVERSITY
Pune

raisoni EDUCATION

A

Project report on

CareerPilotAi

Submitted To :

G H Raisoni International Skill Tech University.
PUNE-412207

For: Bachelor of Computer Applications(Science)

Submitted By :

Prerna Fartade (BC1013)
Shubham Panchal (BC1029)

Semester IV

Year 2025-26

Under the Guidance of :

Prof. Vaishnavi Lokhande

G H Raisoni International Skill Tech University.
PUNE-411007

CERTIFICATE

This is cerftify that Ms. Prerna Fartade and Mr. Shubham Panchal, student of SYBCA, Roll No:BC1013 and BC1029 respectively, has statisfactorily completed their Project "CareerPilotAi" and submitted the report for the academic Year 2026-27

Date:_____

Project Guide
Prof.Vaishnavi Lokhande

_____
Internal Examiner

_____
External Examiner

_____
HOD

_____
Dean

ACKNOWLEDGEMENT

Successful Project is the result of a good team-work Which consists of the not only the Partners Who Put their long and hard work but also those who guided them.

Indeed, a true saying. First of all , I would like to thanks, our Honourable **Vice Chancellor** to give me an opportunity to study in this institute.

I am also grateful to **Ms. Vaishnavi Lokhande** for providing necessary information, valuable cooperation, encouragement and all the technical help during the completion of my project I am also thankful to all faculty members who has supporting me to complete my Project.

Last but not least, we express our gratitude to the almighty without whose blessing nothing is possible.

Signature of Student's:

Prerna Fartade

Shubham Panchal

ABSTRACT

In the modern era, selecting a suitable career path has become increasingly complex due to the wide range of opportunities and rapidly evolving industry requirements. Many students and professionals face confusion in identifying the right career choices, required skills, and preparation strategies.

Traditional **career guidance** methods often lack personalization and fail to provide accurate, data-driven insights. To address these challenges, CareerPilot AI is developed as an intelligent career guidance system that leverages Artificial Intelligence to assist users in making informed career decisions.

CareerPilot AI is a comprehensive platform that integrates multiple functionalities, including career recommendation, resume analysis, interview preparation, and job suggestion. The system collects user data such as interests, skills, academic background, and resumes, and applies AI techniques like Natural Language Processing (NLP) and data analysis to evaluate the information. Based on this evaluation, the system identifies suitable career paths, highlights skill gaps, and provides actionable suggestions to improve employability.

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INTRODUCTION:

In today's fast-changing and **competitive world**, choosing the right career path has become one of the most important yet challenging decisions for students and professionals. Many individuals struggle to identify their interests, strengths, and suitable career options due to lack of proper guidance and awareness.

Traditional career counseling methods are often limited, time-consuming, and not personalized, which leads to confusion and wrong career choices.
With the rapid advancement of technology, Artificial Intelligence (AI) has emerged as a powerful tool that can transform the way career guidance is provided. AI-based systems can analyze large amounts of data, understand user preferences, and generate accurate recommendations. This creates an opportunity to develop intelligent systems that offer personalized and data-driven career advice.

CareerPilot AI is designed as an intelligent career guidance platform that helps users make informed career decisions. The system collects user information such as skills, interests, educational background, and resume data, and uses AI techniques to analyze this information. Based on the analysis, it suggests suitable career paths, identifies skill gaps, and recommends necessary improvements.

PROBLEM DEFINITION :

In today's rapidly evolving and highly competitive world, choosing the right **career path** has become a significant challenge for students and professionals. With the increasing number of career options and changing industry demands, individuals often find it difficult to identify the most suitable career based on their skills, interests, and educational background. Many people lack access to proper guidance and rely on assumptions, peer influence, or incomplete information, which can lead to poor career decisions.

Traditional career counseling methods are often limited, expensive, and not easily accessible to everyone. They usually do not provide personalized recommendations based on an individual's unique profile. As a result, users may remain unaware of the skills required for specific career paths or the gap between their current abilities and industry expectations. This lack of awareness often leads to unemployment, underemployment, or dissatisfaction in chosen careers.

Another major issue is the absence of a unified platform that offers complete career support. Users typically have to depend on multiple tools for different purposes such as resume building, interview preparation, and job searching.

EXISTING SYSTEM

The existing system mainly depends on traditional career counseling and online searches. Users often rely on general advice from teachers, parents, or job portals. There is no personalized guidance based on individual skills and interests. Different platforms are used separately for resumes, jobs, and interview preparation.
Overall, the system lacks integration, efficiency, and intelligent analysis.

NEED FOR NEW SYSTEM

The current **career guidance** methods are not personalized and often lead to confusion. Users need a system that can analyze their skills, interests, and qualifications. There is a need for accurate and data-driven career recommendations. Existing platforms do not provide complete career support in one place. Users require an integrated system for resume building, interview preparation, and job search.

A system is needed to identify skill gaps and suggest improvements. Manual guidance methods are time-consuming and not always reliable. There is a demand for an easily accessible and user-friendly solution. Artificial Intelligence can improve accuracy and efficiency in career guidance.

SCOPE OF SYSTEM :

The scope of the CareerPilot AI system is to provide a smart and efficient platform that assists users in making the right career decisions. The system focuses on analyzing user inputs such as skills, interests, and educational background to generate suitable career recommendations. It aims to simplify the career selection process and reduce confusion among students and job seekers.

The major scope of the system includes:

- Providing personalized career suggestions based on user profile
- Analyzing skills and identifying gaps in user knowledge
- Suggesting required skills and learning paths
- Helping users create professional resumes
- Providing interview preparation through mock tests and questions
- Recommending suitable job opportunities
- The system integrates multiple career-related services into a single platform. This reduces the need to use different applications for resume building, job searching, and preparation.
- The system also improves efficiency by saving time and providing structured guidance to users.

FEASIBILITY STUDY

The feasibility of the system can be explained as follows:

1. Technical Feasibility :

The system is technically feasible as it uses technologies like Node.js, Express, and databases such as MySQL or MongoDB. These technologies are widely available, easy to use, and well-supported. The AI features such as skill analysis and recommendations can be implemented using basic algorithms and APIs.

2. Economic Feasibility :

The system is cost-effective because it uses open-source tools and software. There is no need for expensive hardware or paid licenses. Development and maintenance costs are low, making it affordable for students and small organizations.

3. Operational Feasibility :

The system is user-friendly and easy to operate. Users can easily register, enter their details, and get career suggestions without any technical knowledge. The interface is simple, which ensures smooth interaction and usability.

4. Time Feasibility :

The project can be completed within the given time frame. The modules can be developed step-by-step, making it manageable and efficient to implement within academic deadlines.

HARDWARE AND SOFTWARE REQUIRMENTS :

The development and implementation of the CareerPilot AI system require certain hardware and software components to ensure smooth functioning. Since the system is web-based, it does not need high-end infrastructure and can run on standard computer systems. The following are the basic requirements needed for developing and executing the system.

HARDWARE REQUIREMENTS :

Processor: Intel Core i3 or higher
RAM: Minimum 4 GB (8 GB recommended)
Storage: 500 GB HDD or SSD
System Type: 64-bit Computer
Input Devices: Keyboard and Mouse
Output Device: Monitor
Internet Connection: Required for accessing the system
Network: Stable connection for better performance.

The system requires basic hardware components for development and usage. A computer or laptop with a minimum Intel Core i3 processor is sufficient. At least 4 GB RAM is required for smooth performance. Adequate storage such as HDD or SSD is needed to store project files and data. A stable internet connection is necessary to access and run the web-based system.

SOFTWARE REQUIREMENTS :

he **CareerPilot AI** system is developed using modern software technologies that support web-based applications and intelligent features. These software components are essential for designing the user interface, managing backend operations, and storing user data efficiently. The system mainly uses open-source tools, which makes it cost-effective and easy to maintain.

The major software requirements are as follows:

- **Operating System:** Windows, Linux, or macOS for development and execution
- **Frontend Technologies:** HTML, CSS, and JavaScript for designing the user interface
- **Backend Technologies:** Node.js and Express.js for handling server-side logic
- **Database:** MySQL or MongoDB for storing user data, resumes, and job information
- **Code Editor:** Visual Studio Code for writing and managing source code
- **Web Browser:** Google Chrome or Microsoft Edge for running and testing the application
- **Server Environment:** Node server or XAMPP for hosting the application locally
- **Version Control:** Git and GitHub for managing code and collaboration (optional)
- **APIs:** AI/ML APIs for implementing smart career recommendations

REQUIREMENT ANALYSIS

Requirement analysis is an important phase in system development where the needs and expectations of users are identified and clearly defined. It helps in understanding what the system should do and how it should perform. For the **CareerPilot AI system**, requirement analysis focuses on identifying the features needed to provide effective and intelligent career guidance to users.

1. Functional Requirements :

Functional requirements describe the main operations and features of the system. The system should allow users to register and login securely. It should collect user details such as skills, interests, and education. The system must analyze user data and suggest suitable career paths. It should identify skill gaps and recommend improvements. The system should provide a resume builder feature. It must include interview preparation with practice questions. The system should recommend jobs based on user profile

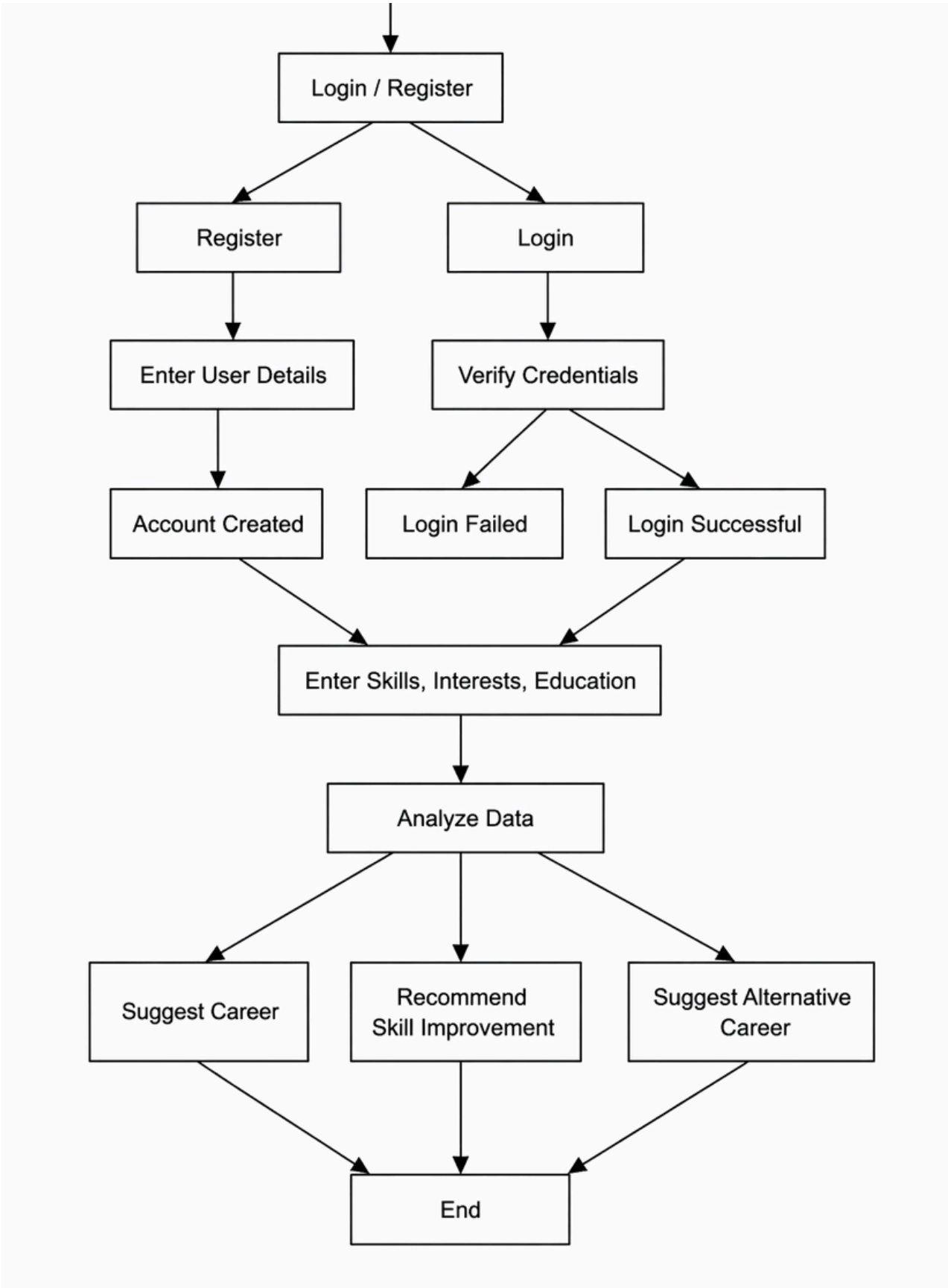
2. Non-Functional Requirements :

Non-functional requirements define how the system should perform. The system should be user-friendly and easy to navigate. It should provide fast and accurate results. The system must ensure data security and privacy. It should be reliable and available at all times The system should be scalable to handle multiple users. It must be compatible with different devices and browsers

DECISION TABLE :

Conditions	Case 1	Case 2	Case 3	Case 4
User has required skills	Yes	Yes	No	No
User interested in field	Yes	No	Yes	No
User has relevant education	Yes	No	Yes	No
Actions				
Suggest Career	Yes	No	No	No
Recommend Skill Improvement	No	Yes	Yes	Yes
Suggest Alternative Career	No	Yes	Yes	Yes

DECISION TREE :



NORMALIZED DATABASE DESIGN (3NF):

USER TABLE :

Column Name	Data Type	Constraints
user_id	INT	PRIMARY KEY
name	VARCHAR	NOT NULL
email	VARCHAR	UNIQUE
password	VARCHAR	NOT NULL

SKILLS TABLE :

Column Name	Data Type	Constraints
skill_id	INT	PRIMARY KEY
skill_name	VARCHAR	NOT NULL

USER SKILLS TABLE :

Column Name	Data Type	Constraints
user_id	INT	FOREIGN KEY
skill_id	INT	FOREIGN KEY

CAREER ROLES TABLE :

Column Name	Data Type	Constraints
career_id	INT	PRIMARY KEY
career_name	VARCHAR	NOT NULL
description	TEXT	

USER_CAREER_PREFERENCE TABLE :

Column Name	Data Type	Constraints
user_id	INT	FOREIGN KEY
career_id	INT	FOREIGN KEY

RESUMES TABLE :

Column Name	Data Type	Constraints
resume_id	INT	PRIMARY KEY
user_id	INT	FOREIGN KEY
resume_content	TEXT	
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTA

JOBS TABLE :

Column Name	Data Type	Constraints
job_id	INT	PRIMARY KEY
career_id	INT	FOREIGN KEY
job_title	VARCHAR	NOT NULL
company_name	VARCHAR	
location	VARCHAR	

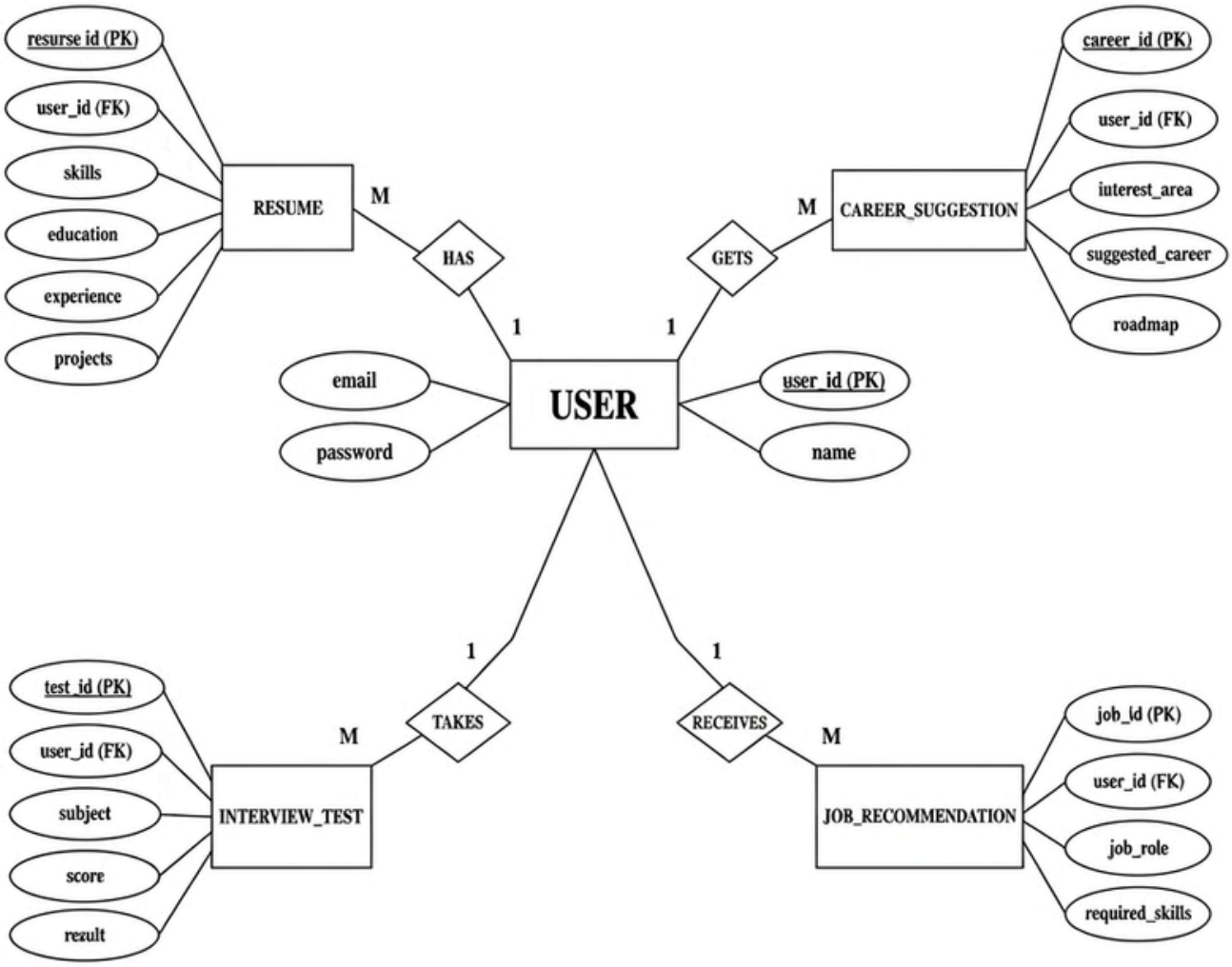
INTERVIEW_TESTS TABLE :

Column Name	Data Type	Constraints
test_id	INT	PRIMARY KEY
user_id	INT	FOREIGN KEY
subject	VARCHAR	(C, C++, Java, HTML,
score	INT	
test_date	DATE	

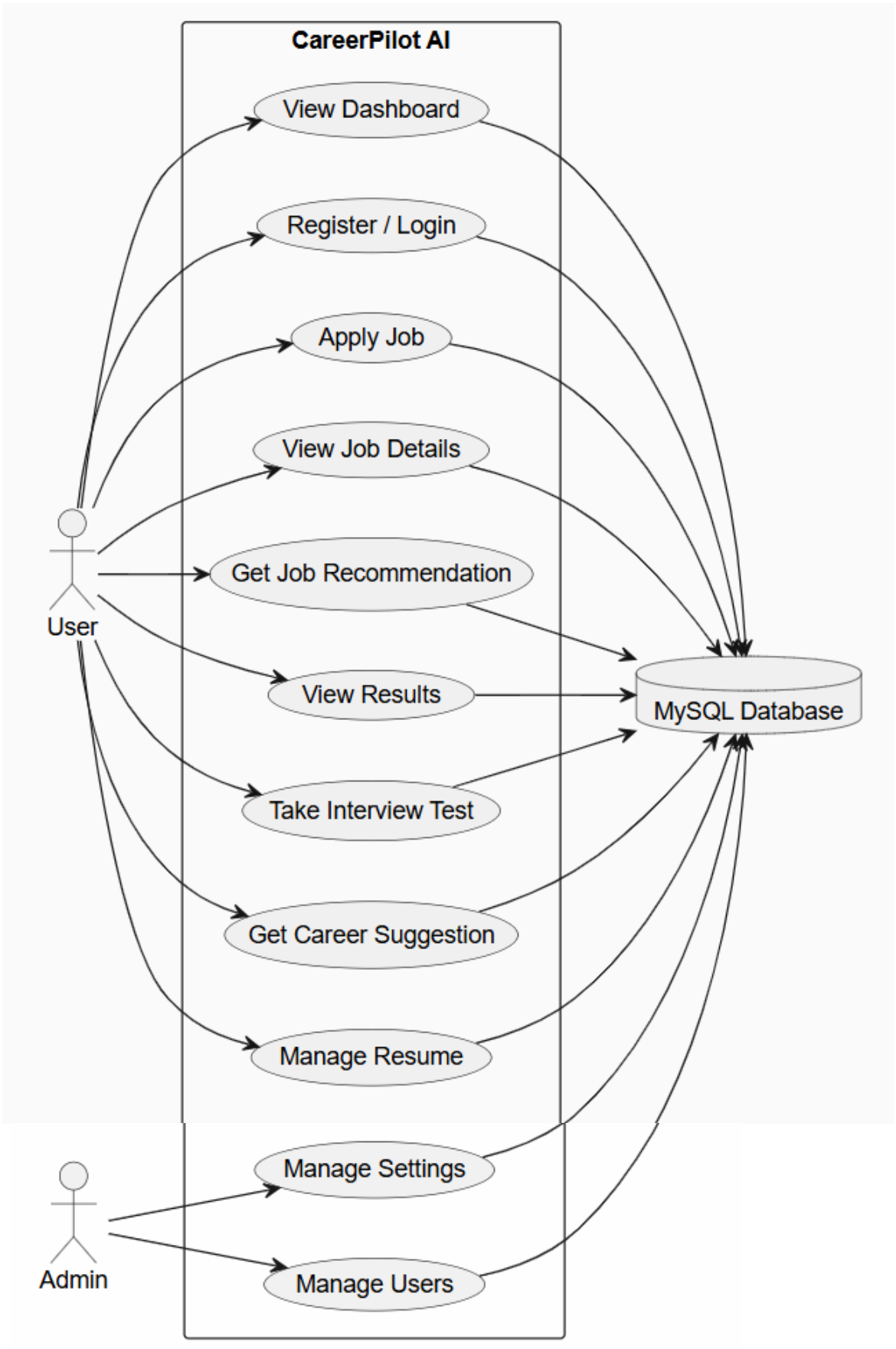
JOB_RECOMMENDATIONS TABLE :

Column Name	Data Type	Constraints
recommendation_id	INT	PRIMARY KEY
user_id	INT	FOREIGN KEY
job_id	INT	FOREIGN KEY
match_score	FLOAT	

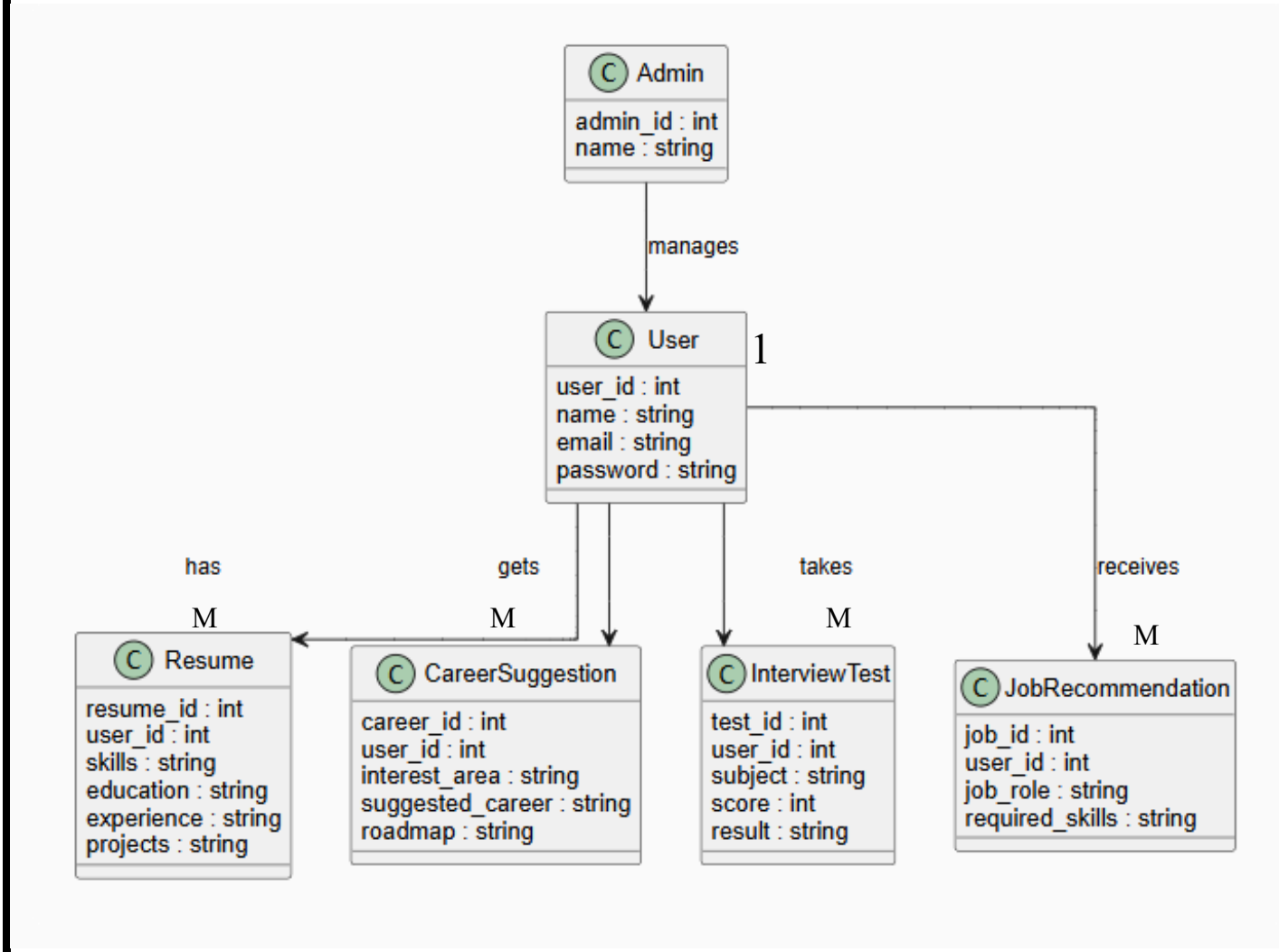
ER DIAGRAM:


resurse id (PK)
user_id (FK)
skills
education
experience
projects
RESUME
M
HAS
1
email
password
USER
user_id (PK)
name
1
GETS
M
CAREER_SUGGESTION
career_id (PK)
user_id (FK)
iuterest_area
suggested_career
roadmap
1
TAKES
M
test_id (PK)
user_id (FK)
subject
score
rezult
INTERVIEW_TEST
1
RECEIVES
M
JOB_RECOMMENDATION
job_id (PK)
user_id (FK)
job_role
required_skills

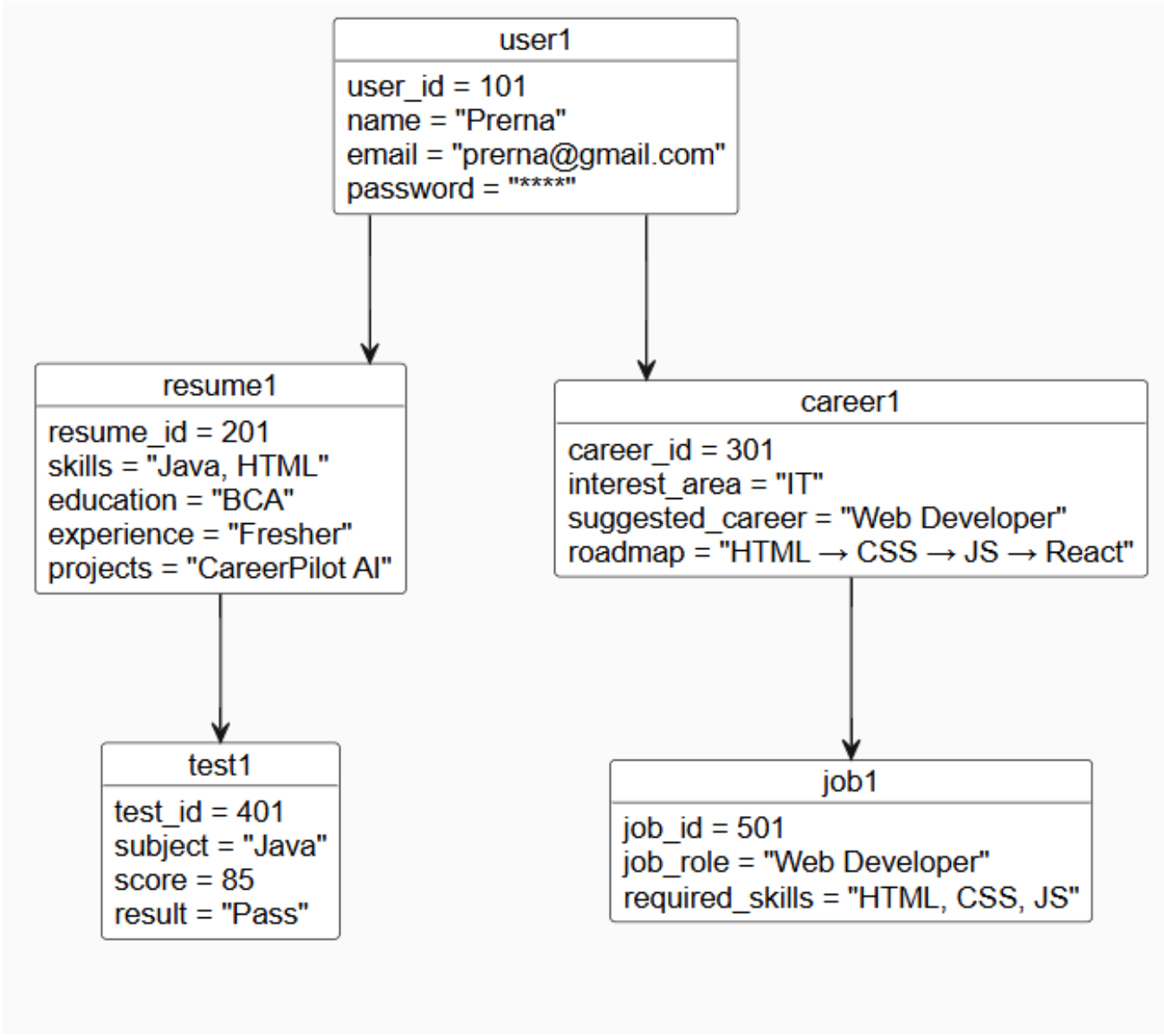
USE CASE DIAGRAM:



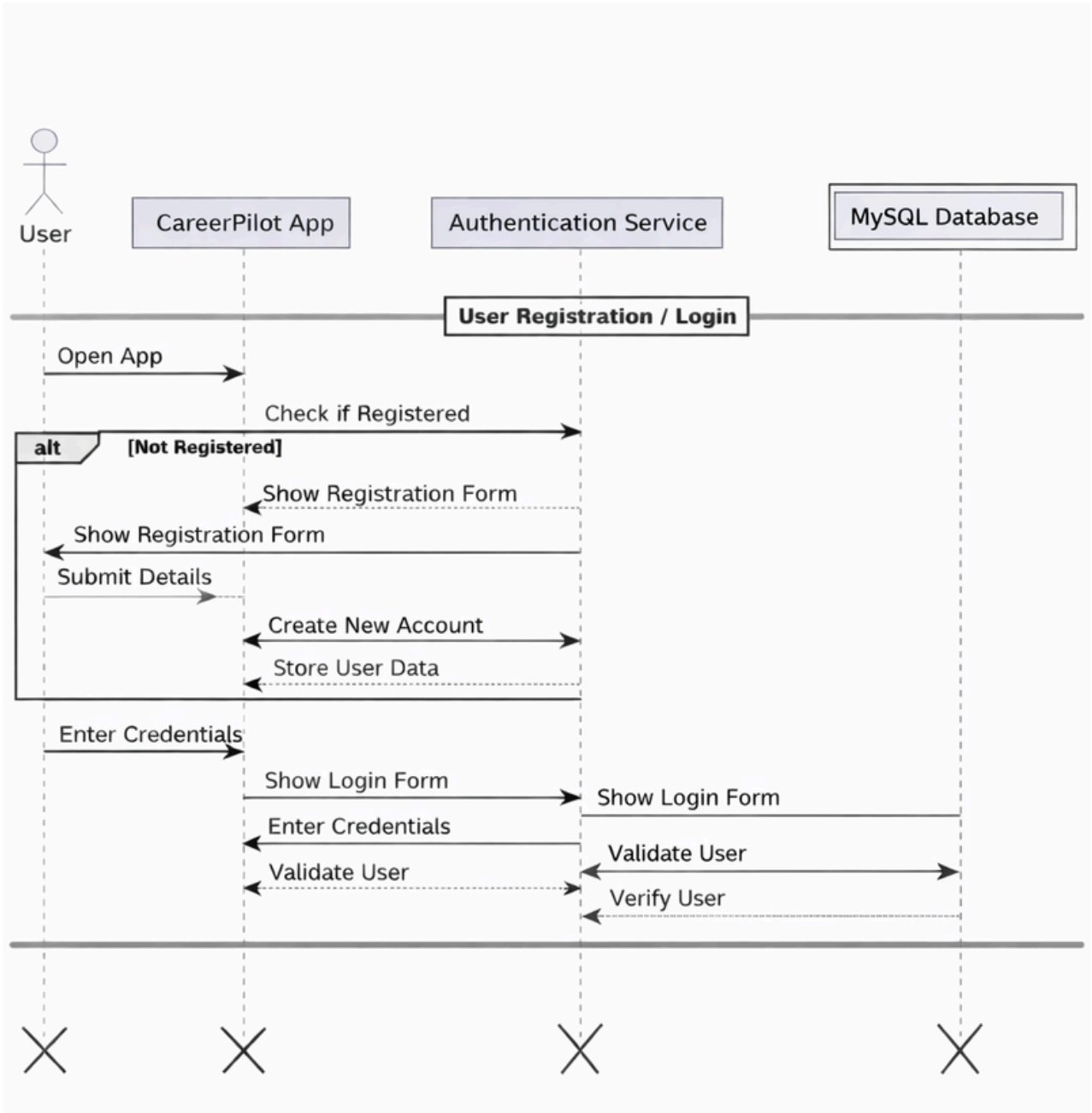
CLASS DIAGRAM



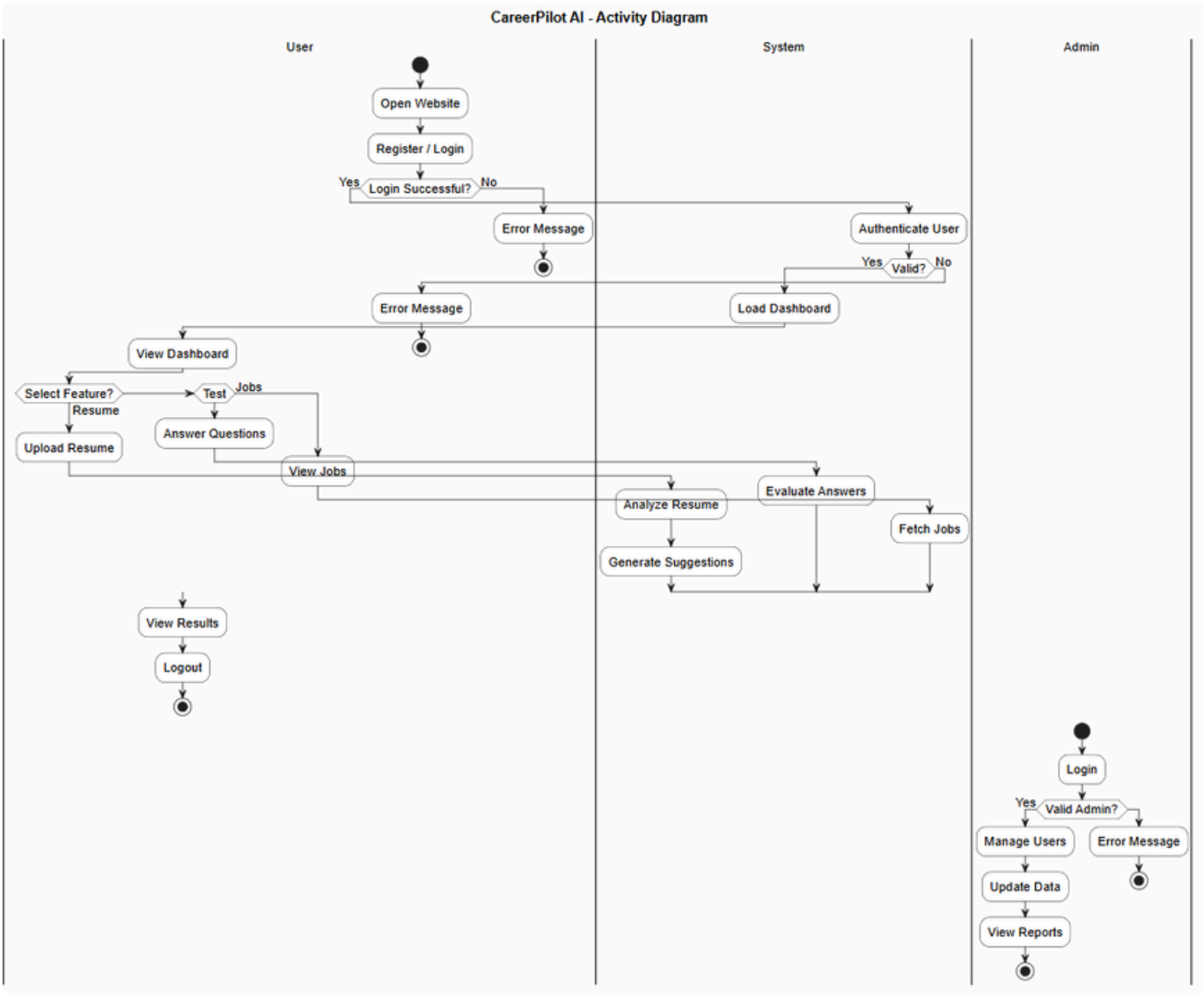
OBJECT DIAGRAM :



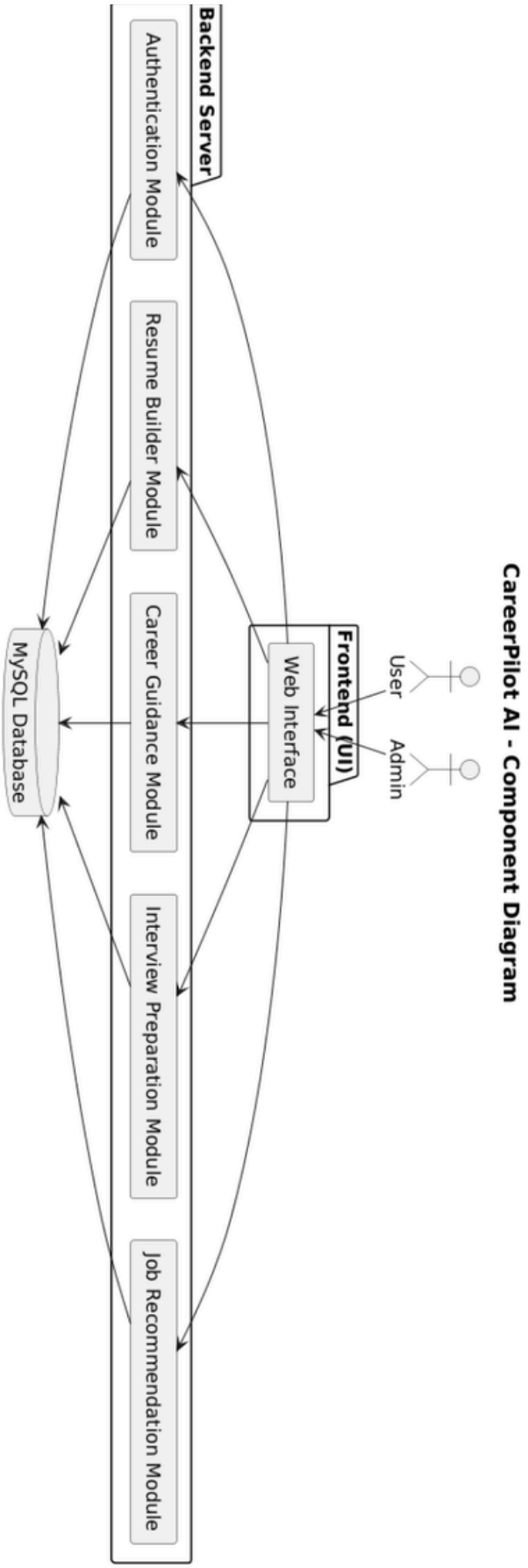
SEQUENCE DIAGRAM :



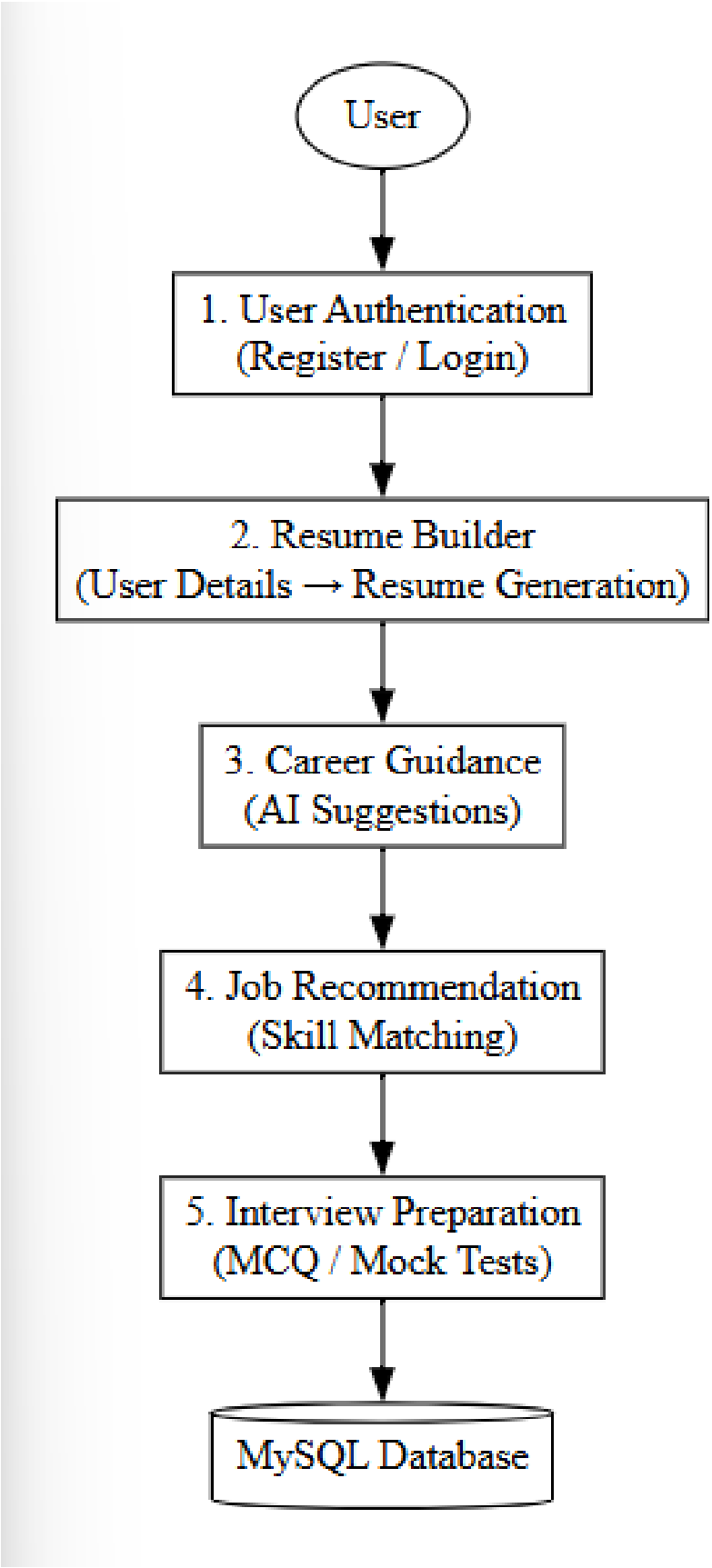
ACTIVITY DIAGRAM :



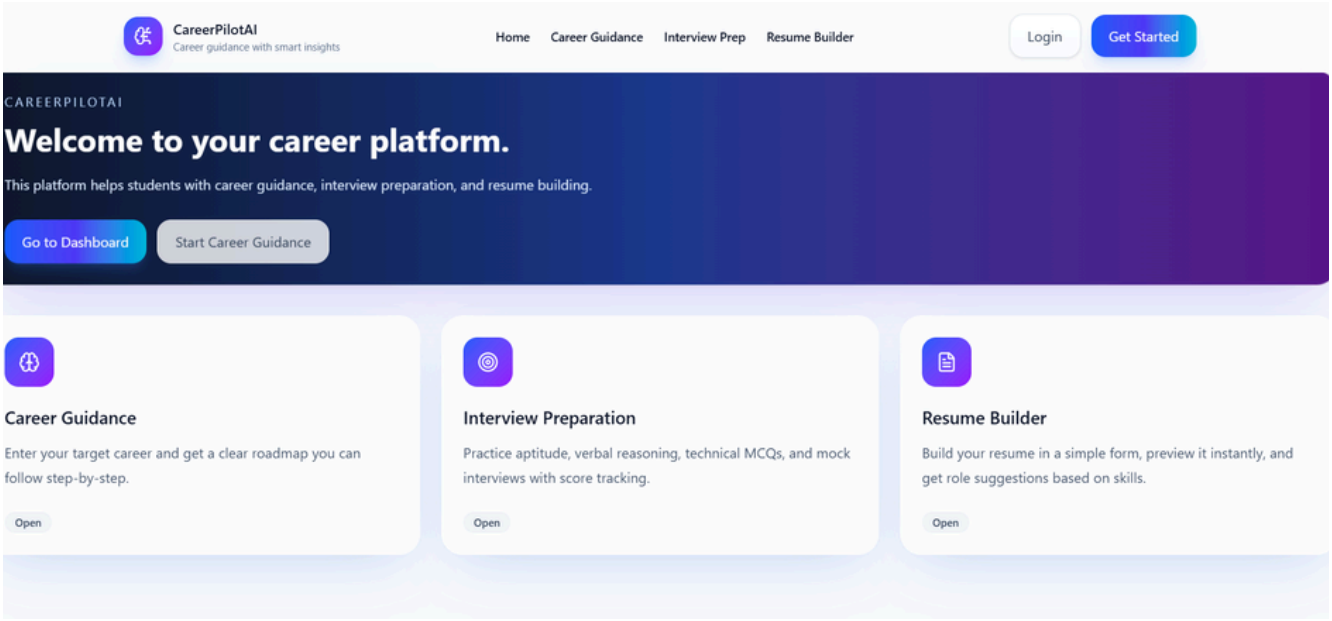
COMPONENT DIAGRAM :

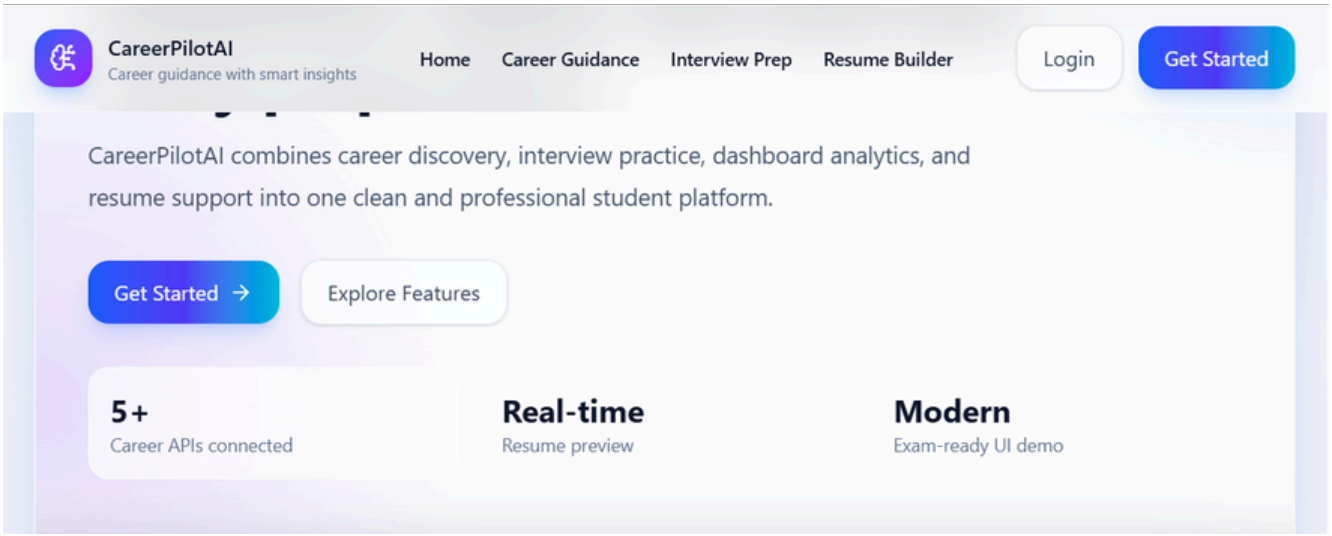


DATA FLOW DIAGRAM :

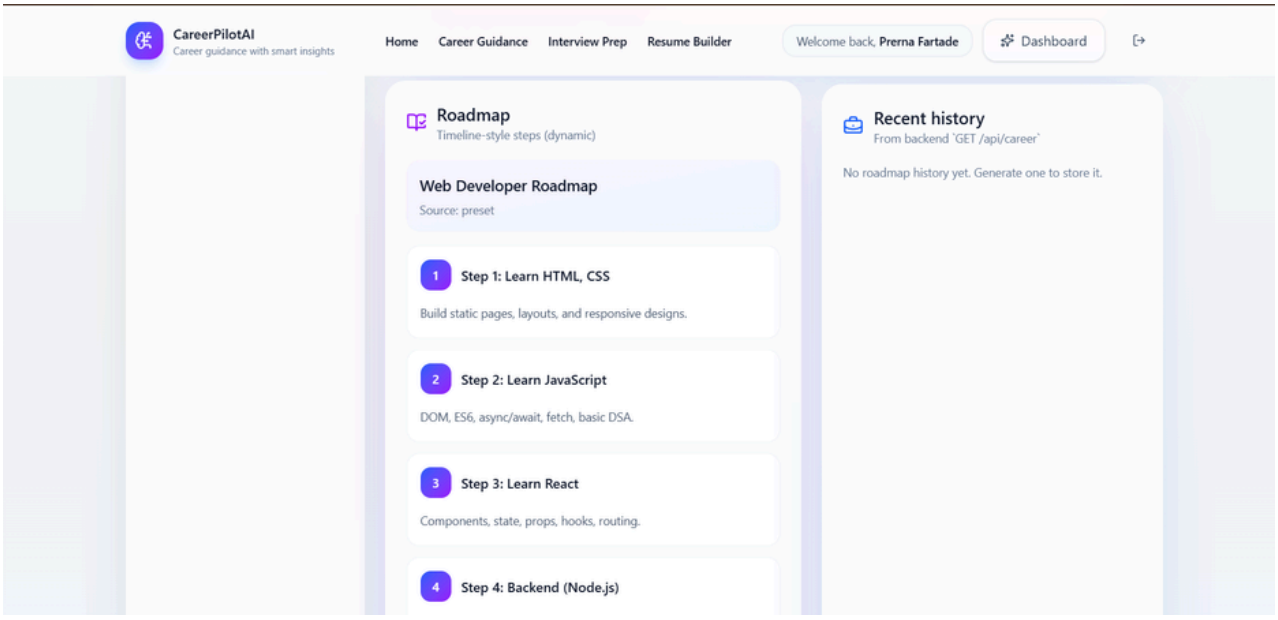


HOME SCREEN :

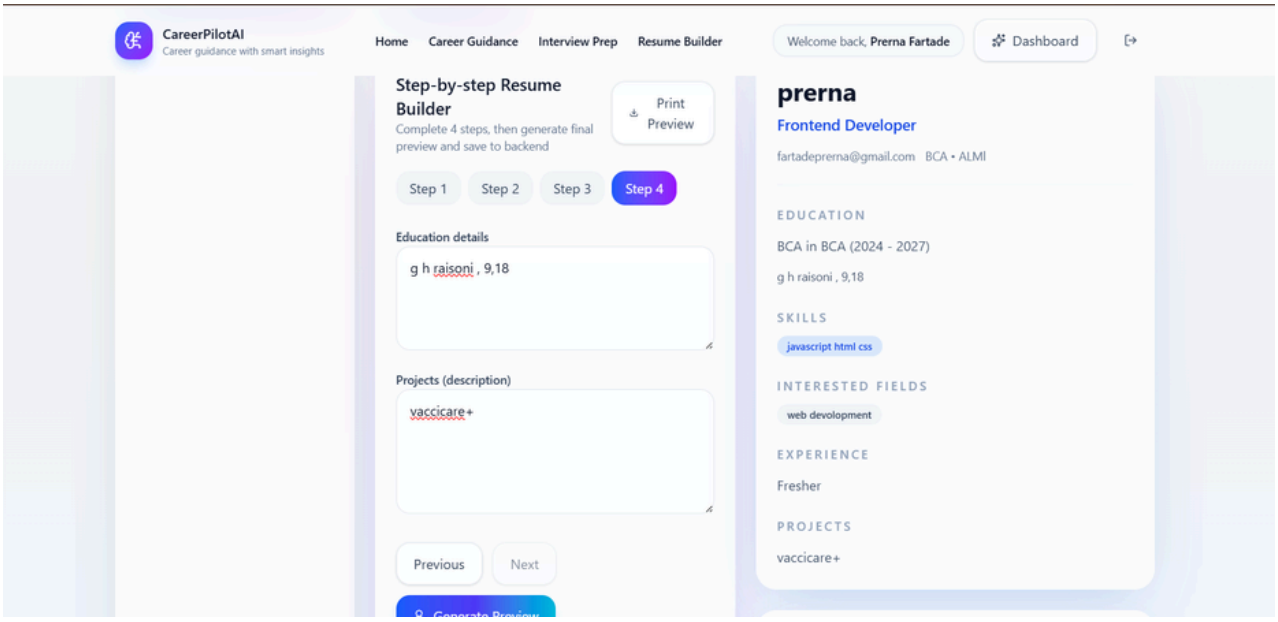

CareerPilotAI
Career guidance with smart insights
Home
Career Guidance
Interview Prep
Resume Builder
Login
Get Started
CAREERPILOTAI
Welcome to your career platform.
This platform helps students with career guidance, interview preparation, and resume building.
Go to Dashboard
Start Career Guidance
Career Guidance
Enter your target career and get a clear roadmap you can follow step-by-step.
Open
Interview Preparation
Practice aptitude, verbal reasoning, technical MCQs, and mock interviews with score tracking.
Open
Resume Builder
Build your resume in a simple form, preview it instantly, and get role suggestions based on skills.
Open

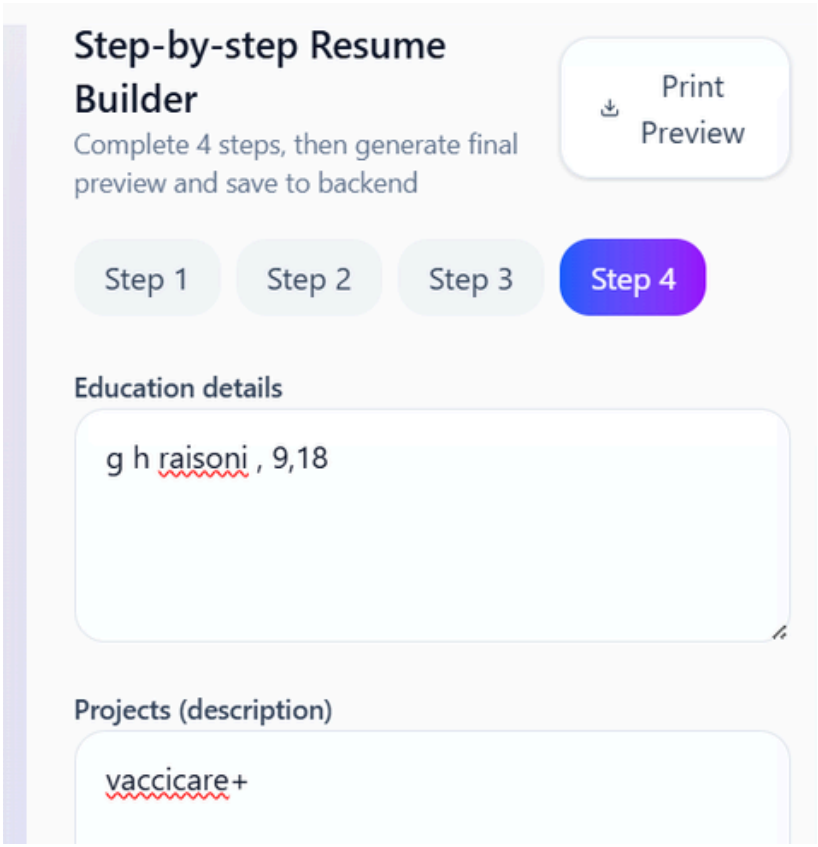

CareerPilotAI
Career guidance with smart insights
Home
Career Guidance
Interview Prep
Resume Builder
Login
Get Started
CareerPilotAI combines career discovery, interview practice, dashboard analytics, and resume support into one clean and professional student platform.
Get Started →
Explore Features
5+
Career APIs connected
Real-time
Resume preview
Modern
Exam-ready UI demo

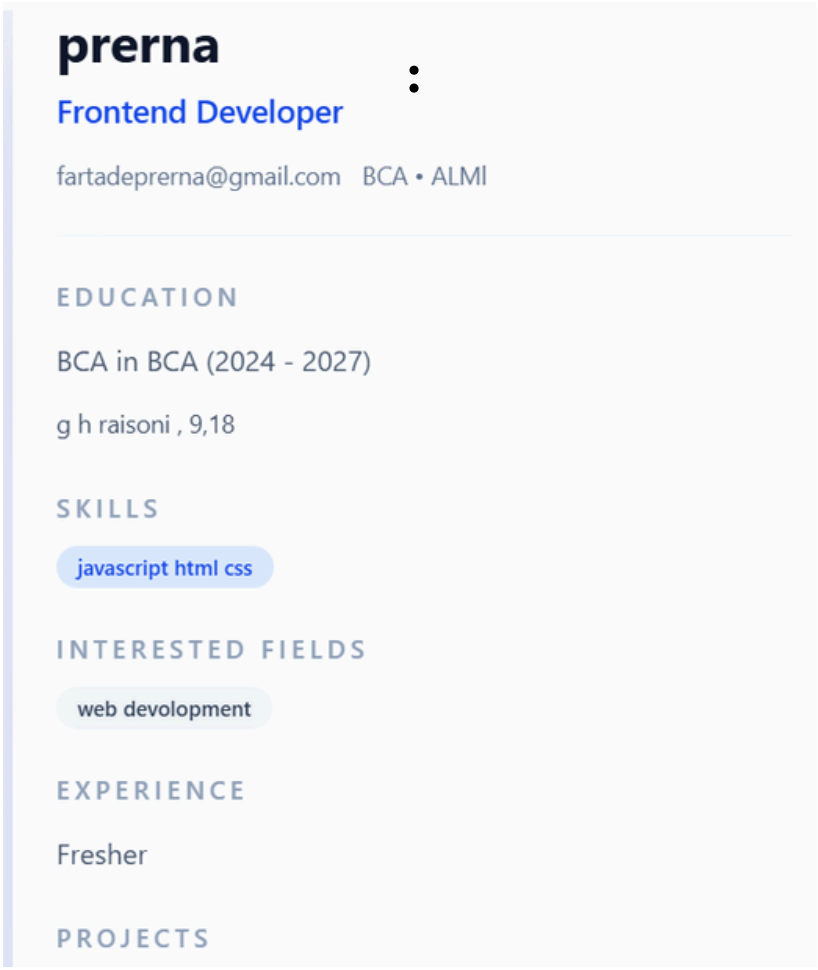
CAREER GUIDENCE MODULE :



RESUME MODULE




Step-by-step Resume Builder
Complete 4 steps, then generate final preview and save to backend
Print Preview
Step 1
Step 2
Step 3
Step 4
Education details
g h raisoni , 9,18
Projects (description)
vaccicare+


prerna
Frontend Developer
fartadeprerna@gmail.com BCA • ALMl
EDUCATION
BCA in BCA (2024 - 2027)
g h raisoni , 9,18
SKILLS
javascript html css
INTERESTED FIELDS
web devolopment
EXPERIENCE
Fresher
PROJECTS

TESTING AND IMPLEMENTATION :

TESTING STRTEGIES:

1. Top-Down Strategy:

Testing begins from the main modules such as the user interface and gradually moves to sub-modules like resume builder, career guidance, and interview preparation.

2. Bottom-Up Strategy:

Individual components like database connection, API functions, and authentication are tested first before integrating them into the main system.

3. Incremental Testing Strategy:

Modules are developed and tested step-by-step. Each new feature is tested after integration to ensure smooth performance.

4. System Testing Strategy:

After integrating all modules, the complete system is tested as a whole to verify overall functionality and performance.

TESTING TECHNIQUES :

1. Black Box Testing:

The system is tested without knowing internal code structure. Inputs and outputs are checked for correctness, such as login validation and career suggestions.

2. White Box Testing:

Internal code structure is tested to verify logic, loops, database queries, and API functions in Node.js backend.

DRAWBACVK AND LIMITATIONS

CareerPilot AI provides useful career guidance, resume building, and interview preparation features, but it also has some limitations. The system is highly dependent on the accuracy and completeness of the data stored in the database. If the data is outdated or insufficient, the career suggestions and job recommendations may not be fully accurate.

Another **limitation** is that the system currently works on rule-based logic and predefined datasets rather than advanced real-time AI models. This reduces the personalization level of career recommendations compared to modern AI-driven platforms. The system also requires an active internet connection and a properly running backend server to function, which may cause issues if the server is down or slow.

Performance may be affected when a large number of users access the system at the same time because it is not fully optimized for high-scale deployment.

PROPOSED ENHANCEMENTS

CareerPilot AI can be further improved by adding advanced features and optimizing the existing system to make it more intelligent, scalable, and user-friendly.
One major enhancement is the integration of Artificial Intelligence and Machine Learning models to provide more accurate and personalized career recommendations based on user behavior, skills, and market trends. The resume builder can also be upgraded with AI-based resume scoring and suggestions to improve content quality.

The **interview preparation** module can be enhanced by adding real-time mock interviews using voice and video interaction, along with a larger dynamic question bank that updates regularly. Job recommendation can be improved by integrating live job **APIs** such as LinkedIn or Indeed to fetch real-time job listings.

Security can be strengthened by implementing JWT authentication, password encryption, and role-based access control. For better performance, caching techniques and optimization of database queries can be applied to handle a large number of users efficiently. Additionally, the **UI/UX** can be improved by making the platform more responsive, interactive, and mobile-friendly.

CONCLUSION :

CareerPilot AI is an innovative web-based platform developed to assist students and job seekers in planning and managing their career paths effectively. In today's competitive environment, many learners face confusion while selecting suitable career options, required skills, and preparation strategies. This system addresses these challenges by providing a centralized solution that combines **career guidance, resume building, job recommendations, and interview preparation tools.**

The main objective of CareerPilot AI is to simplify career decision-making through intelligent suggestions based on user interests, skills, and academic background. It helps users explore different career domains such as software development, web development, data science, and other professional fields.

Along with career suggestions, the system also provides structured roadmaps, which guide users step-by-step about what to learn and how to prepare for their chosen field.

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